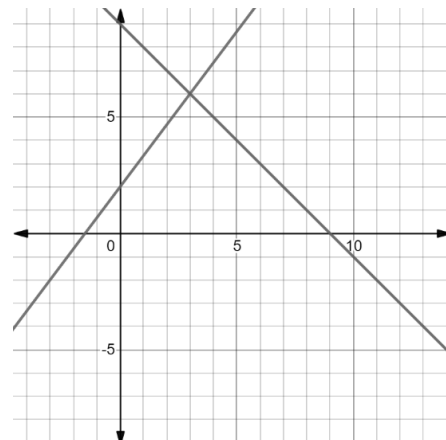


Algebra 1
Unit 5
Lesson 6 Homework

Name Answer Key
Date _____
Period _____

The graph of the system of equations $\begin{cases} y = \frac{4}{3}x + 2 \\ y = -x + 9 \end{cases}$ is shown.



1. To determine the solution to a system of equations, identify the

- ☐ slope.
- ☐ y-intercept.
- ☒ point of intersection.

2. Determine the solution to the system of equations from the graph.

$x = 3, y = 6$
 $(3, 6)$

Verify algebraically that that your solution satisfies both equations in the system.

3. $y = \frac{4}{3}x + 2$
 $(6) = \frac{4}{3}(3) + 2$
 $6 = 4 + 2$
 $6 = 6 \checkmark$

4. $y = -x + 9$
 $(6) = -(3) + 9$
 $6 = -3 + 9$
 $6 = 6 \checkmark$

5. Choose all of the systems of equations that are equivalent to the original system of equations, $\begin{cases} y = \frac{4}{3}x + 2 \\ y = -x + 9 \end{cases}$. Explain how you know the systems are equivalent.

$\begin{cases} -4x + 3y = 6 \\ x + y = 9 \end{cases}$	$\begin{cases} y = \frac{4}{3}x + 2 \\ 4x + 4y = 36 \end{cases}$	$\begin{cases} -\frac{4}{3}x + y = 2 \\ -x - y = -9 \end{cases}$	$\begin{cases} 8x - 6y = -2 \\ y = -x + 9 \end{cases}$
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After rearranging the equations, the slopes and y-intercepts match.

6. Verify your choices in question 5 algebraically.

$-4x + 3y = 6$
 $3y = 4x + 6$
 $y = \frac{4}{3}x + 2 \checkmark$

$x + y = 9$
 $y = -x + 9 \checkmark$

$4x + 4y = 36$
 $4y = -4x + 36$
 $y = -x + 9 \checkmark$

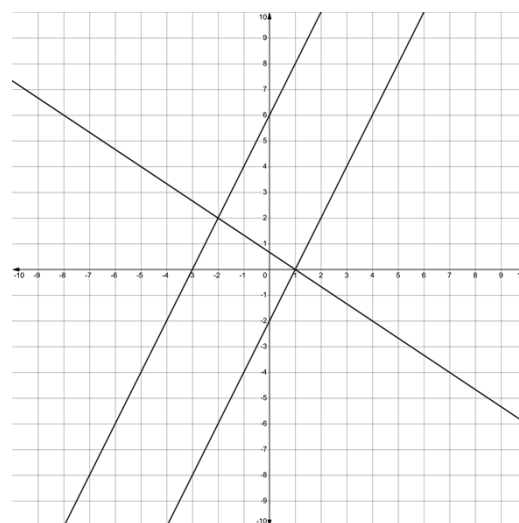
$-\frac{4}{3}x + y = 2$
 $y = \frac{4}{3}x + 2 \checkmark$

$-x - y = -9$
 $-y = x - 9$
 $y = -x + 9 \checkmark$

The graphs of the system of equations $\begin{cases} 2x + 3y = 2 \\ -2x + y = -2 \end{cases}$ and $\begin{cases} 2x + 3y = 2 \\ -2x + y = 6 \end{cases}$ are shown.

7. The solution to the system $\begin{cases} 2x + 3y = 2 \\ -2x + y = -2 \end{cases}$ is $(\underline{1}, \underline{0})$. Verify the solution algebraically.

$$\begin{array}{rcl} 2x + 3y & = & 2 \\ -2x + y & = & -2 \\ \hline 4y & = & 0 \\ y & = & 0 \end{array} \quad \begin{array}{rcl} -2x + 0 & = & -2 \\ -2x & = & -2 \\ x & = & 1 \end{array}$$



8. The solution to the system $\begin{cases} 2x + 3y = 2 \\ -2x + y = 6 \end{cases}$ is $(\underline{-2}, \underline{2})$. Verify the solution algebraically.

$$\begin{array}{rcl} 2x + 3y & = & 2 \\ -2x + y & = & 6 \\ \hline 4y & = & 8 \\ y & = & 2 \end{array} \quad \begin{array}{rcl} 2x + 3(2) & = & 2 \\ 2x + 6 & = & 2 \\ 2x & = & -4 \\ x & = & -2 \end{array}$$

9. Fill in the blanks with words from the word bank to complete the sentence.

The two systems of equations are <u>not equivalent</u> because the two systems result in <u>different</u> solution(s).	Word Bank: ☐ equivalent ☐ equal ☒ not equivalent ☐ not equal ☐ the same ☒ different
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10. Replace one equation in the system of equations $\begin{cases} 2x + 3y = 2 \\ -2x + y = 5 \end{cases}$ so that it is equivalent, but not equal to the system of equations $\begin{cases} 2x + 3y = 2 \\ -2x + y = -2 \end{cases}$.

Answers may vary.

$$y = 2x - 2 \quad \begin{cases} 2x + 3y = 2 \\ y = 2x - 2 \end{cases}$$